

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer (Node.js + TypeScript) building **event-driven systems** for 5+ years — Kafka + NATS Streaming microservices for 4 NGO tenants at CodeBits, real-time Google Pub/Sub sync for a zero-downtime migration of 20+ apps at MTailor, and 2M-record analytical jobs at Dr Wealth. Strong fit for scalable, message-driven backend platforms using TypeScript, Fastify/Express, Kafka or NATS.

CORE COMPETENCIES

TypeScript + Node.js

Event-driven architecture

Kafka + NATS

Pub/Sub messaging

Fastify / Express APIs

Microservices + gRPC

Kubernetes + Docker

Distributed systems

WORK EXPERIENCE

MTailor September 2022 – Present

Software Engineer (US Remote) — reporting to CTO

- Led **zero-downtime migration of 20+ production Node.js apps** from Parse/MongoDB to Firebase/GCP
- Built **event-driven real-time sync via Google Pub/Sub** for two-way MongoDB ↔ Firestore consistency with idempotent consumers and drift detection — high-throughput message processing during live migration
- Implemented a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via a Python script
- Implemented Express Shipping integration, generating **\$40 additional revenue per order**
- Saved the company **\$5,000/month** by migrating all services off AWS

CodeBits January 2020 – July 2021

Software Engineer / Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs in Uganda
- Architected event-driven microservices using Apache Kafka, Docker, Kubernetes, NATS Streaming, and gRPC** for FIDA Uganda — multi-tenant pub/sub with typed contracts shared across 4 NGO tenants
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda
- Developed a cross-platform mobile app for LASPNET using React Native and Expo
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC

Dr Wealth August 2021 – January 2022

Software Engineer (Singapore Remote)

- Optimized a PWA delivering real-time stock prices from a Firebase backend
- Built ad-hoc jobs to **query over 2 million Firestore records**, keeping prices fresh under SLA from Morningstar APIs
- Extended backend APIs using **Firebase Cloud Functions and Express.js** hosted on Heroku

Mind2matter January 2022 – July 2022

Software Engineer (US Remote)

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines

MTailor March 2022 – July 2022

Contractor (US Remote)

- Built a Node.js migration tool reading from MongoDB and writing to Firestore
- Wrote SDK documentation and trained Ops team on Firebase Dashboard and backup procedures

EDUCATION

B.Sc. Computer Engineering · Makerere University, Kampala 2018 – 2022

SKILLS

Proficient: TypeScript, Node.js, JavaScript, Kafka, NATS, gRPC, Pub/Sub, microservices, Kubernetes, Docker, Express, MongoDB, SQL, GCP (Firestore, Cloud Functions, GCS), Python, Git, Linux

Intermediate: Go, AWS (S3, EC2, EBS), Rust, React, React Native, Fastify (familiar), Redis (familiar)

Languages: English (native), Kiswahili